



Enhanced glaucoma detection using ensemble based CNN and spatially based ellipse fitting curve model

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Abstract

Glaucoma is one of the leading causes of blindness for people over the age of 60. For the automatic glaucoma detection using fundus images, several approaches have been recently developed. However, the extraction of optic cup boundary considered as critical work due to the blood vessels interweavement and also the significant challenges in accurate localization of affected optical disc based on size of the cup which is either too small or large. To overcome this challenges efficient technique needs to be considered. To accurately detect the glaucoma and correspondingly the optic cup OC and optic disk OD boundary the ensemble based CNN classification with spatially based ellipse fitting curve model (SBEFCM) has been proposed. In this study, the glaucoma or diabetic retinopathy DR classification has been performed by the ensemble CNN classification. The optic disc and optic cup boundary detected by new spatially weighted ellipse fitting model SBEFCM. For the proposed SBEFCM, which enhances and prolongs the multi-ellipse fitting technique. The ensemble based CNN classification is performed for various datasets such as LAG, RIM-ONE and local retinal glaucoma dataset. Previously the input fundus image is pre-processed and further blood vessel segmentation performed to avoid the interweavement with surrounding tissues and blood vessels. The new SBEFCM accurately detected the ratio of OC and OD boundary prediction which are depicted and by using the ensemble CNN classification, various performance metrics such as sensitivity, specificity, accuracy and AUC values has been identified for glaucoma detection and by comparison the proposed ensemble CNN outperforms the existing methods.

Keywords Glaucoma/diabetic retinopathy detection · Spatially based ellipse fitting curve · Ensemble CNN · Optic cup OC · Optic disk OD

1 Introduction

Computer aided diagnostics enhances the diagnostic speed and assisted in locating the specific area. Diabetic retinopathy (DR) is occurs at the back of the eye (retina) due to the blood vessels damage caused by prolonged Diabetes mellitus. New, abnormal blood vessels are manufactured by retina. Neo-vascular glaucoma are then occurs if the new blood vessels grows on the iris, fluid flows is then closing off in the eye and eye pressure is raised. The major challenge while detecting the Glaucoma is it doesn't show any symptoms/pains and also the vision is normal at an early stage. During the advanced stage it is noticeable when the patient

has lost 70% of his/her vision. Hence for early glaucoma detection regular screening of eye is necessary. Basically for DR detection the Fundus photography is widely used by ophthalmologist (Kumar et al. 2019).

Generally for medical diagnosis pre-processing, feature selection and disease classification are followed. The challenges seen in glaucoma detection is the accurate localization of affected optical disc and based on size of the cup which is either too small or large. There are several clinical features exists in the diabetic retinopathy such as micro aneurysms (MAs), haemorrhages, hard and soft exudates shown in Fig. 1. Hence, the feature extraction is considered as the substantial part in detecting the glaucoma (Salamat et al. 2019). The DR classification is termed as proliferative diabetic retinopathy (PDR) and non-proliferative diabetic retinopathy (NPDR) based on the ischemic change and vessel degeneration. The PDR is the DR's advanced stage whereas NPDR is sub categorised into mild (MA's

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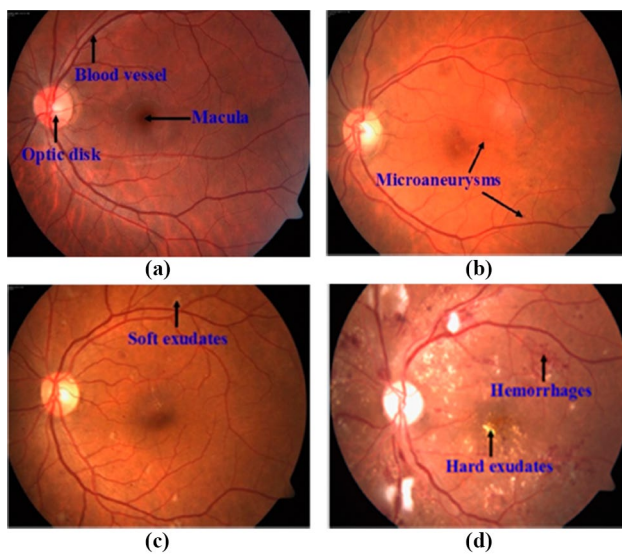


Fig. 1 (Qureshi et al. 2019) **a** Anatomical features of retina, **b–d** mild, moderate and severe glaucoma

occurrence), moderate and severe stages. Extraction of optic cup boundary is considered as critical work due to the blood vessels interweavement (Qureshi et al. 2019; Qureshi 2015).

From the existing studies, the challenges seen are the extraction of optic cup boundary considered as critical work due to the blood vessels interweavement. Generally for medical diagnosis pre-processing, feature selection and disease classification considered as significant challenges and in case of glaucoma detection, accurate localization of affected optical disc based on size of the cup which is either too small or large considered as great risk.

The proposed ensemble classification with spatially based ellipse fitting curve model (SBEFCM) are overcome these challenges and the major contribution of this study includes,

- To perform optic disc OD and optic boundary OC accurately by implementing the SBEFCM algorithm.
- To detect the diabetic retinopathy/glaucoma in retinal images by using the ensemble based classification.
- To compare the proposed method with various existing methodologies and datasets to show the effectiveness of the current study.

2 Related works

The following section describes the literature review related with suitable existing approaches for glaucoma detection and the existing challenges.

By using retinal images, efficient techniques such as particle swarm optimization improved ensemble deep neural networks have used in the existing studies for the optic disc

OD segmentation. The single network bias has been overcome by the ensemble segmentation technique (Zhang and Lim 2020). The two critical issues seen in optical nerve head is variant spatial layout of vessels and small spatially sparse optic cup boundaries solved by the spatial-aware joint segmentation method (Liu et al. 2019). Due to the large variation, highly weak contrast among optic head areas and heavy overlap the multi-task collaborative learning MCL-Net has been developed for the quantification of optic nerve head multi-indices. However in some cases it is complex to identify the OC/OD boundary with indistinct and glaucoma fundus (Zhao and Li 2020). A Multi label deep learning framework namely M-Net used for OC and OD segmentation issues by generating the one stage multi-label model. Higher accuracy resulted for vertical disc and cup compared to horizontal (Fu et al. 2018). Generally, for medical image analysis the deep learning approaches are not used for glaucoma evaluation due to its small data size and hence this issue overcome by REFUGE-retinal fundus glaucoma challenge consists of large data with 1200 fundus images. Further the two basic tasks such as OD/OC segmentation and classification of glaucoma have been performed (Orlando et al. 2020). The DENet-disc aware ensemble network based deep learning model has been further developed for local optic disc boundary incorporation and global fundus image integration related to the deep hierarchical context. In addition to that the DENet gives glaucoma detection precisely even without segmentation (Fu et al. 2019; Veena et al. 2021).

It is possible that the optic disc and cup are not directly segmented and hence the optic disc and cup of minimum bounding boxes has been detected and evaluates as adorned ellipses inside the boxes. Joint RCNN namely Faster RCNN has been used for this process of optic disc and cup boundary boxes detection (Jiang et al. 2019). For the multi-scale fundus image representation a new framework based Weakly Supervised Multitask Learning WSMTL based CNN implemented and the model trained based on binary diagnosis such as pixel-level evidence map, normal or glaucoma, diagnosis prediction and segmentation mask (Zhao et al. 2019; Vaghjani et al. 2020). The second leading cause of blindness globally is Glaucoma which is not reversed and it is loss of vision caused by Glaucoma. For high quality Glaucoma screening, Yanbao—user-friendly app developed based on the clinical parameters (Guo et al. 2018). Within the retinal images the complex blood vessel structure and severe intensity inhomogeneity have been considered as severe issues and it overcome by locally statistical active contour model (LSACM) incorporates probability information based on multi-dimensional feature (Gao et al. 2019). In fundus images, the segmentation and localization of optic disc is considered as significant step for early retinal disease onset detection like glaucoma, diabetic retinopathy and macular degeneration. For this process a novel CNN used for OD

segmentation (Bhatkalkar et al. 2020; Raja et al. 2020). In retinal fundus images, the basic two-stage Mask R-CNN has been developed for optic nerve head localization and optic disc/cup segmentation. This study enhanced the detection with the help of new fine-tuning approach by cropping output combined with original training image using various scales. Vertical cup to disc proportion measured for glaucoma diagnosis (Almubarak et al. 2020; García et al. 2020).

Through a systematic classification pipeline, the regional classification problem is modelled from the OD segmentation problem in retinal fundus images. By textural and statistical properties the image regions are characterized in a classification framework and hence this kind of technique is potential against the OD localization corresponds to multifaceted challenges (Rehman et al. 2019). The DenseNet integrated with fully convolutional network namely FC-DenseNet designed for the semantic segmentation. Thus, OD and OC automatic segmentation performed in this study and effectiveness resulted. Also the automated classification results enhanced by vertical cup to disc ratio with horizontal cup to disc ratio and however, the diagnosis results can be improvise by utilizing the OD's complete profile (Al-Bander et al. 2018). OD segmentation challenges are occurs due to the vessels presence and it has been expected to overcome by pre-processing techniques. This techniques used morphological operations such as histogram equalization or closing and opening. However compared with OD, OC segmentation challenges shows complexity because of the optic cup interweavement with surrounding tissues and blood vessels. Thus enhanced segmentation techniques are necessary for accurate glaucoma diagnosis (Thakur and Juneja 2018). Large number of images should be used to improve the accuracy in glaucoma diagnosis. Because full potential is not resulted and also only few filters have trained. It leads to not giving assurance of capturing all variables of retinal images from the given images (Zilly et al. 2017). By extracting the specific features, feature extraction method is used based on intensity and texturing by using modified multi-text on structure descriptor and classify abnormality in brain MRI and SVM has been performed (David and Jayachandran 2018). The suitable classifier algorithm needed for possible outcome which determine the relationships among the set of attributes (David 2019; David and Jeyachandran 2016). Further multiclass lesion classification system has been developed using the RBF kernel based SVM and hybrid colour image structure descriptor in retinal images used to classify the lesions into normal and abnormal classes (Jayachandran and David 2018; David and Jayachandran 2020).

2.1 Paper organisation

The following Sect. 2 describes the proposed ensemble CNN classification with SBEFCM and followed by the proposed

method evaluation and comparison with existing methods illustrated in Sect. 3. Finally the paper concluded in Sect. 4.

3 Methodology

In this study, the glaucoma classification has been performed by the ensemble CNN classification. The optic disc and optic cup boundary detected by new spatially weighted ellipse fitting model. Previously the input fundus image is pre-processed and further blood vessel segmentation performed to avoid the interweavement with surrounding tissues and blood vessels.

The overall flow of the proposed system is shown in Fig. 2.

3.1 Pre-processing of the fundus image

The original input fundus image is in the form of low intensity false regions and the denoising technique applied which includes the image transformation into domain where the noise is easily recognized and eliminated by applying threshold. To reconstruct the noise free image the inverted transformation is applied. Hence to remove the false regions, de-noising the image is necessary which preserves the vessel edges.

3.2 Vessel segmentation

By using the fixed threshold scheme, vascular structure is extracted. For segmentation the decision parameter is fixed to some constant in this scheme.

If $I_{x,y}$ is a grayscale image, then by fixed threshold the binary image $bw_{x,y}$ is obtained as,

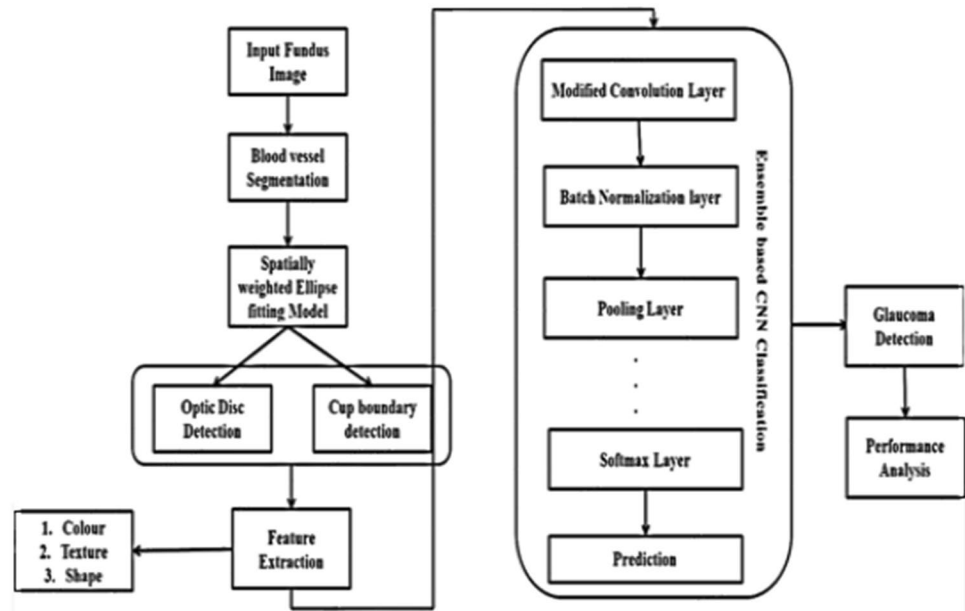
$$bw_{x,y} = \begin{cases} 1 & \text{if } I_{x,y} \geq th \\ 0 & \text{if } I_{x,y} < th \end{cases} \quad (1)$$

For all given dataset images the threshold parameter value is fixed and shows variation from dataset to dataset. Based on accuracy evaluation the threshold is selected by several threshold values selection on the individual datasets. On the given dataset, the final threshold produces greater accuracy.

3.2.1 Post-processing

To increase the system performance, three post processing operations are used such as elimination of spur pixels, filling gaps and area filtering. To remove small isolated and unwanted regions which is not belong to vessel structure, the area filtering operation is used. Based on 8-way pixel connectivity of first labelling of pixels into components, it is obtained using connected component analysis (CCA). To

Fig. 2 Proposed flow for glaucoma detection and analysis



classify them as vessel or non-vessel, each labelled component area is measured. If the area is less than A those regions are non-vessel regions else they are vessel. At vessel structure edges, the spur pixels are removed. To fill small gaps, at last the structuring element of disk type based morphological closing operation is applied.

3.3 Spatially based ellipse fitting curve model (SBEFCM)

For the proposed SBEFCM, which enhances and prolongs the multi-ellipse fitting technique. The binary image I which characterizes the 2D shape. A p is the pixel of I fits to the background $B_g(I_p = 0)$ or the foreground $F_g(I_p = 1)$. The 2D shape area is represented as,

$$Area_{2D} = \sum_{p \in F} I_p \quad (2)$$

In addition to that, the s ellipses E_n set by single area is $|E_n|$. A bw_{im}^p binary image is represented and hence, $bw_{im}^p = 1$ at t points are includes to any of the ellipses E_n or else $bw_{im}^p = 0$. Further, the α_E coverage of the 2D shape is defined by provided set of ellipses E as,

$$\alpha_E = \frac{1}{Area_{2D}} \sum_{p \in F} I_p bw_{im}^p \quad (3)$$

Fundamentally, 2D points shape of percentage α_E are depends on ellipses E . Let the sum of the areas of all ellipses denoted by $|E|$, $|E| = \sum_{n=1}^s |E_n|$. Let the area covered by all ellipses refer as C_{area} is expressed as,

$$C_{area} = \sum_{p \in I} bw_{im}^p \quad (4)$$

In the case $C_{area} \leq |E|$ with equality holding are defined as pairwise disjoint. It is due to the two overlapping ellipses, $|E|$ counts the intersection area dual times whereas C_{area} is not count the intersection area.

The multi-ellipse fitting method evaluates the set of parameters E^* of s ellipses E_n^* and hence α_{E^*} is denoted previously on the Eq. (3) based on the constraints of equal area is maximized. On constraint, it holds $|E^*| = Area_{2D}$. As an alternative, in SBEFCM we maximize the— α_E , coverage of shape with E^* —is set of ellipses in which its C_{area^*} is likely so near to $Area_{2D}$. It can be expressed as,

$$E^* = \underset{E}{argmax} \alpha_E \quad s.t. C_{area} = Area_{2D} \quad (5)$$

With the purpose to evaluate the optimal number ellipses s , the trade-off among the shape coverage α_E and the complexity of model is optimized by using the AIC-Akaike Information Criterion. On the source of circles radii which are on the 2D skeleton shape centered these are calculated. The minimization quantity of overall possible ellipses amounts s by AIC based selection criterion model is expressed as,

$$AIC(E, O) = c \ln(1 - \alpha_E) + 2s \quad (6)$$

Subconsciously, it obtains a good balance among the shape coverage in increased manner which realised as more ellipses used for certain shape approximation with increased model complexity. The model selection process and shape complexity measure are invariant for translation

and shape rotation which are affected slightly due to the changes in scale by the resolution/quantization problems.

3.3.1 SBEFCM algorithm

The SBEFCM operated as same as multi-ellipse fitting method which are précised as,

3.3.1.1 Extraction of skeleton Initially, the 2D shape of medial/skeleton axis S_k is calculated which gives significant information on ellipse parameters which in turn the original shape is approximated. To the minor boundary deformations, the medial axis skeletonization is sensitive related to the significant skeleton branch resulted from the small perturbation of shape boundary. To ease this issue, shape thinning with closing morphological filter employed instead of the medial axis under skeletonization.

3.3.1.2 Ellipse hypotheses initialization SBEFCM defines the circles set CC used as ellipse hypotheses initially. On S the center of these circles are placed and by centers of minimum distance their radii are described from the shape contour. In decreasing order based on radius, focused on circles for CC inclusion. At first $CC = \emptyset$. In every CC focused, the circle is presented if the below certain threshold the already selected circles are overlapped. The SBEFCM complexity and cardinality of CC reduced the radius of the circle lesser than 3% of radius which is maximum are unnoticed. The hypotheses of initial number of circles generates greater number with upper bound of ellipses fitting.

Ellipse hypotheses evolution:

For calculating the fixed number parameters of ellipse in E with 2D shape α_E best coverage, the Gaussian mixture model expectation maximization (GMMEM) technique. This is obtained by two steps,

1. Assigning shape point to ellipses.
2. Estimating the ellipse parameters.

3.3.1.3 Assigning shape point to ellipses Assigned point t to an ellipse E_n if t is inside that ellipse. Generally,

$$F(t, E_n) \leq 1.0 \quad (7)$$

where,

$$F(t, E_n) = \frac{\|t - k_n\|}{\|t' - k_n\|} \quad (8)$$

From Eq. (8) the origin of ellipse E_n is k_n , the intersection of the line is t' connecting t and k_n with E_n ellipse and the 2D vector length termed as $\|\bullet\|$. Hence $F(t, E_n) = 1.0$ for t points place on E_n boundary.

The ellipses may overlap, with more than one ellipses a shape point t may be associated.

3.3.1.4 Estimating the ellipse parameters The ellipse parameters E_n are updated directly by the moments of second order points related to previous ellipse shown in assignment step.

3.3.1.5 Optimal number of ellipses solving Based on AIC criterion described above the various models are evaluated which balances trade-off among approximation error and model complexity. From large described automatically set, the AIC criterion minimized by SBEFCM which decreases the number of ellipses. The number of ellipses reduces since there is no lesser bound on AIC. Until the ellipses set contains single ellipse this process proceeded. An ellipse pairs for merging is selected and represented as candidates in every iteration. For adjacent ellipses merging, multi-ellipse fitting method is considered. However, for merging SBEFCM is considered any ellipses pair. Lowest AIC is resulted from the finally merged pair. The SBEFCM reported as final solution from all possible models exhibited as the one with reduced AIC.

3.3.1.6 Spurious solutions forbidden For E_n ellipse, the $O_r(E_n)$ -overlap ratio is defined as points percentage that are union of the remaining ellipses overlapping expressed as,

$$O_r(E_n) = \frac{|E_n \cap (\cup_{y=1, y \neq n} E_y)|}{|E_n|} \quad (9)$$

$O_r(E_n)$ is greater that a predefined threshold $T_{pre} = 95\%$ forbidden which is considered as a model part.

An ellipse E_n with overlap $O_r(E_n)$ is greater that a predefined threshold $T_{pre} = 95\%$ is forbidden from being concentrated as a part. Figure 2 is an example for this. E_2 ellipse shows greater $O_r(E_2)$ value, hence, small contribution to image foreground coverage related with outcome includes only E_1 and E_3 . From the final outcome it has excluded. The below Fig. 3 considered as an example in which the T_{pre} value is set approximately which handled the results of segmentation. The application of SBEFC with $T_{pre} = 40\%$ and $T_{pre} = 95\%$ exhibited as left, under segmentation and right, perfect near segmentation respectively.

The multi-ellipse fitting method computational complexity is same as SBEFCM which is similar to $O(r2f)$ in which the number of foreground pixels is f and $r = |RR|$ which defines the no. of circles establishes the ellipses hypotheses initially illustrating the 2-D shape.

In the proposed approach, the first step is the Bradley segmentation technique and hole filling method. By

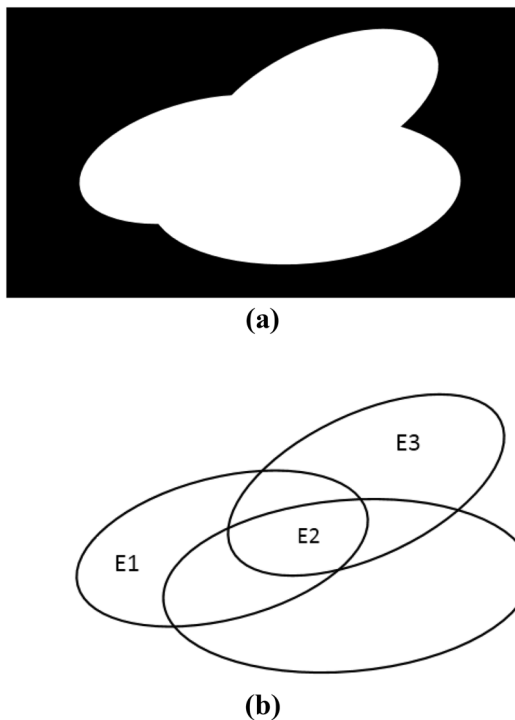


Fig. 3 **a** 2D shape **b** E1, E2 and E3 of three ellipses in which E2 shows large overlap and thus it doesn't mainly contributed for shape coverage increased

imposing the image smoothing technique better results obtained, for e.g. prior to segmentation with Gaussian filter $\sigma_G = 2$ achieved. Locally adaptive image threshold is measured by Bradley technique under the local, statistics of first order image around every pixel. Against the illumination changes this method shows potentiality and it overtakes the Otsu's technique which shows robust illumination variations. The Bradley's technique drawback is that the segmentation in background with local greater brightness recognised mistakenly as cells. The two-shape and one appearance based constraints are introduced to decrease the false positive drawback.

- *(Shape) Area constraint* Each disc's expected area go beyond the minimum threshold T_δ . Hence specifically small segments are rejected. To avoid this rejection which are visible partially T_δ is imposed not to area of measured object but rather than the expected area approximation which in turn measured as area of circle that is fitted best.
- *(Shape) Roundness constraint* The roundness measure are used for disc like circular or elliptic like objects which reject the complex shape objects that diverge considerably. The Roundness measure R shows the resemblance of the shape of an object closeness and it is expressed as,

$$R = \frac{4\pi\delta}{\vartheta^2} \quad (10)$$

From Eq. (10), where δ and ϑ defines the object's area and perimeter. For the perfect circle, the maximum value of 1 is taken by Roundness R . For the proposed approach the roundness must be $R > 0.2$ is needed for a region represented as disc.

Constraints of intensity:

The above described shape constraints are sufficient to forbidden various false positives. The other intensity based constraints which are not associated to shape-based constraints in which false positives are forbidden. This kind of constraints perception is that the intensity distribution within remaining cells same as intensity distribution within cell. The Bhattacharyya distance is employed for measuring the distance among the two intensity distribution based on normal distribution assumption. Two distributions are considered such as d_1 and d_2 , μ_n is the means and σ_n^2 is the variances, where $n \in \{1, 2\}$, the $B_D(d_1, d_2)$ is the Bhattacharyya distance expressed as,

$$B_D(d_1, d_2) = \frac{1}{4} \left[\ln \left(\frac{\sigma_1^2}{4\sigma_2^2} + \frac{\sigma_2^2}{4\sigma_1^2} + \frac{1}{2} \right) + \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2} \right] \quad (11)$$

By using the proposed constraints the four false positives are forbidden.

3.4 Classification

3.4.1 Proposed ensemble framework

The proposed ensemble framework has been seen as ConvNet which is simulated more than one time refer as pipelines. With various parameter settings each trained on class labels subset. The subset of classes or labels referred as subset of dataset. This characteristically termed as the training subsets generated are mutually exclusive. For large datasets with ConvNet training, the main benefit seen as mutifold while training on subset of classes is the minimized time taken for training which not includes the requirement of high end GPUs.

The following Fig. 4 displays all the components initially from transfer learning process whereas from the trained AlexNet the new model received the initial weights. From all classes the reference images are selected and similar images grouping constitutes by Hierarchical agglomerative clustering. Mutually exclusive subsets generated based on this grouping which has given to the network after the training stage.

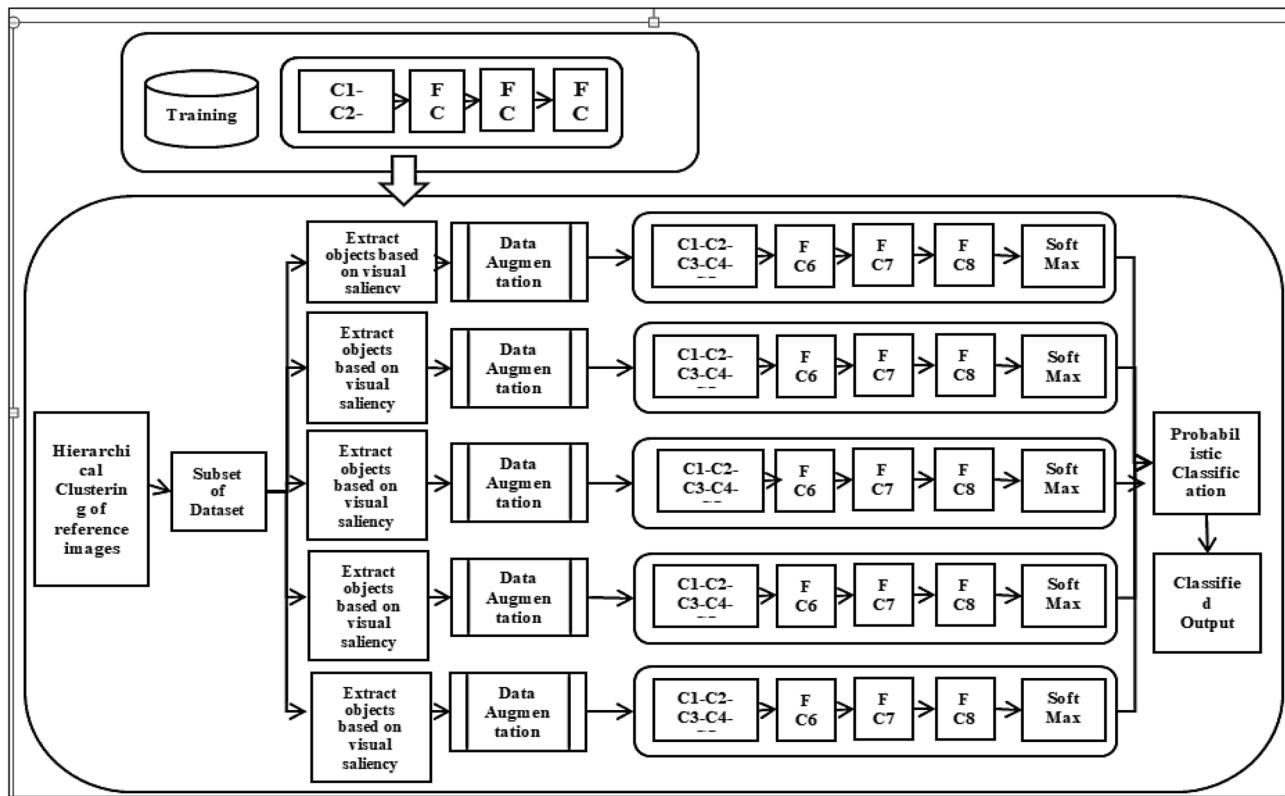


Fig. 4 Proposed ensemble framework for classification of glaucoma in retinal images

3.4.1.1 Transfer learning From AlexNet model on ImageNet the mid-level feature representations are learned effectively which are shifted for new network training. In first seven layers the generalized features or mid-level features are captured. It is described as from the primary Conv layer to next fully connected layer. In our model the learned weights of these layers are used and it is not updated during the training also it kept constant. The classifier of the source task and final fully connected layer FC8 are more particular to the ImageNet and thus it is forbidden. Followed by softmax classifier and novel FC8 are included which are retrained then.

3.4.1.2 Ensemble training In ensemble model, the classes after the grouping are provided as inputs to every pipeline. Generally on GPUs the ConvNets are trained. However the new proposed model is trained without GPU with the input

parameter settings such as 10 testing data and 50 training data which are trained for 15 epochs totally with 10 batch size and learning rate as 0.03. The rectified linear units ReLU is considered as an activation function.

$$f_a(x) = \max(0, x) \quad (12)$$

If the input image is $I(x, y)$ and the filter is $w_f[p, q]$ then the fundamental convolution operation is provided as,

$$c_g(x, y) = \sum_{p=0}^m \sum_{q=0}^n w_f[p, q] * I(x + p, y + q) \quad (13)$$

For the comparative study, the ensemble framework with similar number of pipelines is trained on the whole dataset with variations in parameters. Top-1 and Top-5 errors measured. Both metrics are decreased in training phase.

Algorithm- 1**Input-** I : Set of Images**Output-** net : Trained network**Procedure** Proposed-ensemble (I)

1. With pre-trained AlexNet, Initialize the weights.
2. From each class, Select the reference image
3. Calculate Similarity score matrix S_M .
4. Set of clusters $Cl \leftarrow$ (stratified clustering (S_M))
5. **For** each dataset $h \in Cl(h)$ **do**
6. Saliency Extraction(h);
7. $net \leftarrow Train(h)$.
8. **End for**
9. **Return** trained model net .

End Procedure.

It is based on assumption that provided test image, the pipeline has learned the features precisely will identify it with greater probability related with pipelines of incorrect classification.

Algorithm 2- CNN Testing**Input-** Image selected from respected dataset**Output-** The object is labelled with predicted score**Procedure** CNN_Test

Each dataset images- Load the saved net models
 Replace the softmax loss layer end by softmax classifier.

For the given image each dataset computes the score using similar pooling and convolution operations during the process of training.

From each pipeline, Find maximum of scores. Let it $score(t)$, where the dataset represented as t

$$Score_{final} \leftarrow score(t)$$

Return the respected label $Score_{final}$.**End Procedure.**

4 Results and discussion

4.1 Dataset description

For glaucoma detection, the LAG database and RIM-ONE database used in this proposed study. The dataset link for LAG dataset obtained from <https://computervisiononline.com/dataset/1105138869> and RIM-ONE dataset obtained from <https://www.kaggle.com/nonatosales/rimone>. The LAG database constitutes 11,760 fundus images with respect to 4878 positive and 6882 negative glaucoma images shown in Fig. 5. The number of individuals with several images are tabulated in below figure. From the Beijing Tongren Hospital and Chinese Glaucoma Study Alliance (CGSA) the fundus image are obtained. By using the dataset, the treatment and diagnosis capacity of vision-threatening eye diseases with Glaucoma has been improved further by using the proposed techniques.

The performance of the proposed method on the other public database namely RIM-ONE is concentrated on the open nerve evaluation (ONH) constitutes of 160 high resolution images. By using high accurate segmentation algorithms appropriate abnormalities detected.

Finally, the local retinal glaucoma dataset images are obtained from various public and private resources in various hospitals. The dataset constitutes of totally 1338 retinal images. Every image is appropriate to one of four classes such as Normal, Early, Moderate and Advanced Glaucoma. This kind of dataset consists of 79% images which appropriate to one class and to remaining three classes the 21% if images are presented. With no glaucoma/normal images it is 79.08% images, early glaucoma images of 8.86%, moderate glaucoma of 5.98% and advanced glaucoma of 5.98% images are presented.

In case of LAG database, 10,658 retinal fundus images from seven different sources were found and downloaded. They were randomly divided into 7106 for training and 3551 for validation. Glaucoma prevalence was 24%. Using the validation set, we created 50 random sets of 1000 images with a 5% glaucoma prevalence. On these sets, the classifier

SUMMARY OF OUR LAG DATABASE.

Source Database	Images No.	Positive No. (%)	Individuals No.	Age, Mean (SD)	Female No. (%)	Camera
CGSA	7,463	2,749 (36.8)	6,441	54.1 (14.5)	55.8	Topcon, Canon, Carl Zeiss
Beijing Tongren Hospital	4,297	2,129 (49.5)	3,706	52.8 (16.7)	49.7	Topcon, Canon
Full LAG	11,760	4,878 (41.5)	10,147	53.6 (15.3)	53.6	Topcon, Canon, Carl Zeiss

Fig. 5 LAG database constitutes retinal images(Li et al. 2019)

reached a detection rate of 84.1% (CI 95 = $\pm 1.1\%$) and 95.8% (CI 95 = $\pm 0.2\%$) specificity. In case of RIM-ONE dataset, 160 fundus images are obtained. 130 images are utilized for training and 32 images for testing. 25 images are meant for validation process.

4.2 Performance metrics

Performance analysis is carried out by calculating the values of various performance measures like accuracy, specificity, sensitivity and area under curve (AUC).

4.3 Accuracy

Accuracy is the measurement of standard values and known as the weight arithmetic mean and it is considered as the inverse value of the precision value. The accuracy can be calculated based on the formula given below.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (14)$$

4.4 Sensitivity

Sensitivity is the measurement of the ratio of true positives to the sum of true positives and false negatives. It can also be known with the name of true positive rate. The value of sensitivity is calculated based on the following formula

$$\text{True positive rate} = \frac{TP}{TP + FN} \quad (15)$$

4.5 Specificity

Specificity defines the measures of the proportion of negatives that are correctly identified. It is represented as a false-positive rate or true negative rate and is the ratio of true negatives to the sum of true negative and false positive. It is calculated by using the formula mentioned below,

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (16)$$

4.6 Area under curve (AUC)

Area under curve (AUC) is measured through trapezoidal approximation. The AUC can be expressed as,

$$\text{The area under curve} = \frac{1}{2} \times \left[\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right] \times 100 \quad (17)$$

From Table 1, the pre-processing in case of glaucoma detection is exhibited. Error rate is low and the proposed method shows its efficiency.

4.7 Comparative analysis of glaucoma detection

The severity in glaucoma detection evaluated comparatively with various existing methods. From the LAG, RIM-ONE the comparison performed. The following Table 2 depicted the sensitivity, specificity, accuracy and AUC results of the proposed glaucoma detection compared with existing methods. The proposed method performed with LAG dataset shows 99.51% accuracy, 98.14% sensitivity, 95.65% specificity and 0.9725 AUC whereas for RIM-ONE dataset it shows 93% accuracy, 94.44% sensitivity, 91.30% specificity and 0.941 AUC in the glaucoma detection of retinal images.

Table 1 Pre-processing in case of glaucoma detection

	MSE	PSNR	SSIM	NCC	NPCR
Average filter	0.00012	29.1142	0.9754	0.9981	0.9250
Wiener filter	0.0007	31.6703	0.9923	0.9990	0.9941
Median filter	0.025	26.0736	0.9963	0.9968	0.9681
Proposed	0.0008	15.324	1.000	1.000	0.9965

Table 2 Comparative analysis of proposed and existing techniques (Li et al. 2019) in terms of various metrics with LAG and RIM-ONE dataset

Database	Method	Accuracy	Sensitivity	Specificity	AUC
LAG	Deep CNN (Li et al. 2019)	89.2	89.4	89.9	0.953
	A deep learning system for the classification (Li et al. 2019)	89.7	91.4	88.4	0.96
	AG-CNN (Li et al. 2019)	96.2	95.4	96.7	0.983
	Proposed	99.51	98.14	95.65	0.97
RIM-ONE	Deep CNN (Li et al. 2019)	80	69.6	87	0.831
	A deep learning system for the classification (Li et al. 2019)	67.8	67.4	68.1	0.731
	AG-CNN (Li et al. 2019)	85.2	84.8	85.5	0.916
	Proposed	93	94.44	91.30	0.941

Table 3 Comparative analysis of proposed and existing techniques (Aamir et al. 2020) in terms of accuracy with local retinal glaucoma dataset

Existing/proposed methodologies	Accuracy
CNN (Aamir et al. 2020)	83
Support vector machine (Aamir et al. 2020)	87
CNN (Aamir et al. 2020)	90
Semi-supervised transfer learning for CNN (Aamir et al. 2020)	92.4
Feed-forward neural network (Aamir et al. 2020)	92
OCT probability map using CNN (Aamir et al. 2020)	95.7
Glaucoma-deep (CNN, DBN, softmax) (Aamir et al. 2020)	99
AG-CNN (Aamir et al. 2020)	95.3
ML-DCNN (Aamir et al. 2020)	99.39
Proposed	99.51

The proposed Ensemble classification with SBEFCM performed better for all metrics compared with other existing method deep CNN as shown in table. The sensitivity shows higher value than specificity value. Larger AUC resulted compared with existing methods. Compared with RIM-ONE dataset, the LAG database shows higher values in terms of all metrics. The sensitivity metric is significant since other indicators such as field of vision and intra-ocular pressure confirmed glaucoma diagnosis. It shows that high generalization ability resulted. The existing method performance highly degrades as it shows the over fitting issue.

From the below Table 3, the classification results of the proposed method accuracy and their comparison with other existing methods are shown by using local retinal glaucoma datasets (Aamir et al. 2020). The proposed ensemble based

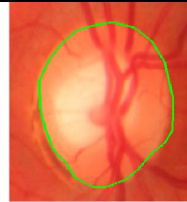
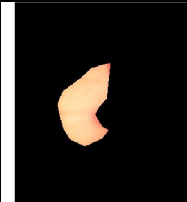
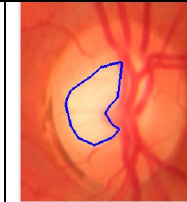
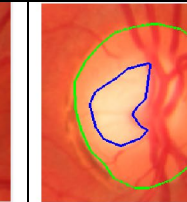
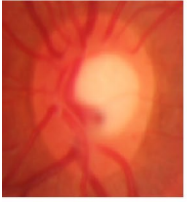
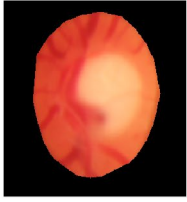
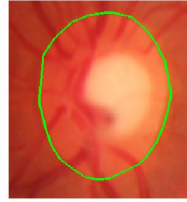
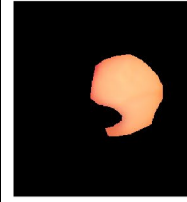
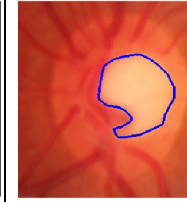
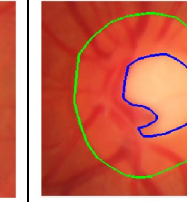
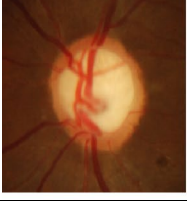
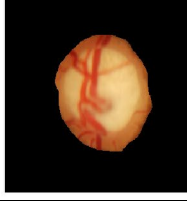
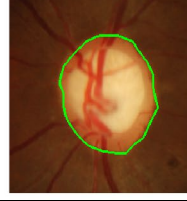
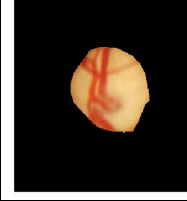
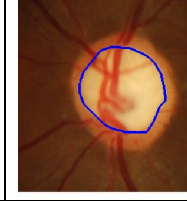
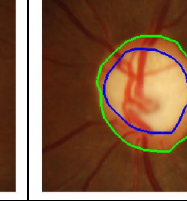
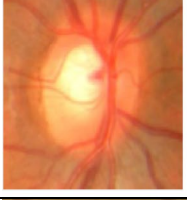
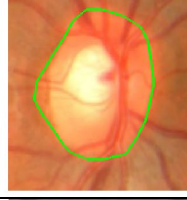
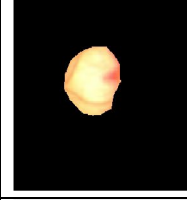
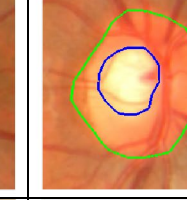
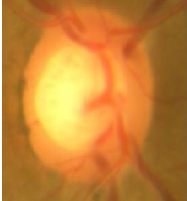
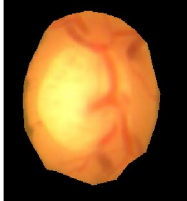
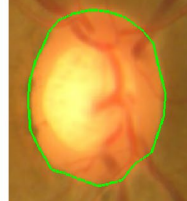
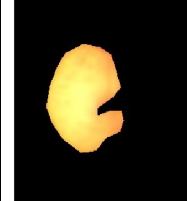
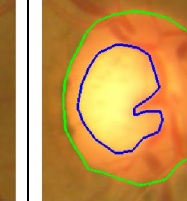
classification with SBEFCM algorithm shows better statistical values compared with the existing methodologies for glaucoma detection. Thus accuracy is 99.51% shows in glaucoma detection obtained using proposed technique.

From the below Table 4, it clearly shows the input sample images, the optic disc OC and optic boundary OD which are detected precisely. By using the proposed SBEFCM the ratio of OC and OD boundary prediction are accurately performed.

5 Conclusion

In this proposed study, a new segmentation method namely SBEFCM for accurately detect the optic disk and optic cup boundary has been performed. Further the severity in glaucoma has been detected using the ensemble based CNN classification for accurate prediction. The SBEFCM operated as same as multi-ellipse fitting method and in which spurious solutions are rejected. For the comparative study, the ensemble framework with similar number of pipelines is trained on the whole dataset with variations in parameters. The datasets used for analysis are the LAG, RIM-ONE and local retinal glaucoma detection dataset. The experimental results shows that the ratio of the OC and OD boundary predicted and depicted in the study. The performance of glaucoma detection using the proposed method are compared with several existing methods exhibited that the proposed method shows better statistical values in terms of accuracy, specificity, sensitivity and AUC. The LAG dataset shows higher values compared with RIM-ONE dataset. For future enhancement, the glaucoma detection is classified based on this proposed algorithm in terms of its mild, moderate and severe NPDR.

Table 4 Optic disc OD and optic cup OC boundary detection by proposed SBEFCM method

Input Image	Optic Disk Detection	Optic Disk Boundary Detection	Cup Detection	Cup Detection	Optic Disk and Cup Prediction
					
					
					
					
					

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Code availability Not applicable.

Declarations

Conflict of interest I confirm that this work is original and has either not been published elsewhere, or is currently under consideration for publication elsewhere.

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